



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Larry Castro - SpaceY
August 5, 2025



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of Methodologies

The project employed a data-driven approach, combining data collection from SpaceX's API and web scraping of Wikipedia pages. Data wrangling and exploratory analysis (EDA) were conducted using Python, SQL, and visualization tools like Matplotlib and Plotly. Interactive maps were built with Folium to analyze launch site performance, while predictive modeling used classification algorithms (Logistic Regression, Decision Trees, SVM). Model tuning was performed via GridSearchCV, and a Plotly Dash dashboard was developed for dynamic data exploration.

- Summary of Results

Key findings revealed KSC LC-39A as the most successful launch site (76.9% success rate), with payload mass showing no direct correlation to outcomes. Decision Trees achieved the highest prediction accuracy (87.8%). Temporal trends indicated steady improvement post-2013, peaking in 2019. SQL queries highlighted operational insights, such as NASA's total payload mass (45,596 kg). Interactive visualizations confirmed geographic and technical factors as critical to launch success.

Introduction

- **Project background and context**

The commercial space era is already underway, with companies such as Virgin Galactic, Rocket Lab, Blue Origin, and especially SpaceX leading the sector. The latter has achieved milestones such as sending spacecraft to the International Space Station, deploying the Starlink satellite constellation, and conducting crewed missions. SpaceX's competitive advantage lies in the reuse of the first stage of the Falcon 9 rocket, which significantly reduces launch costs (USD 62 million compared to over USD 165 million from other providers).

The project is set within this context of innovation and competition, where SpaceY, a new company founded by Elon Musk, seeks to position itself against SpaceX.

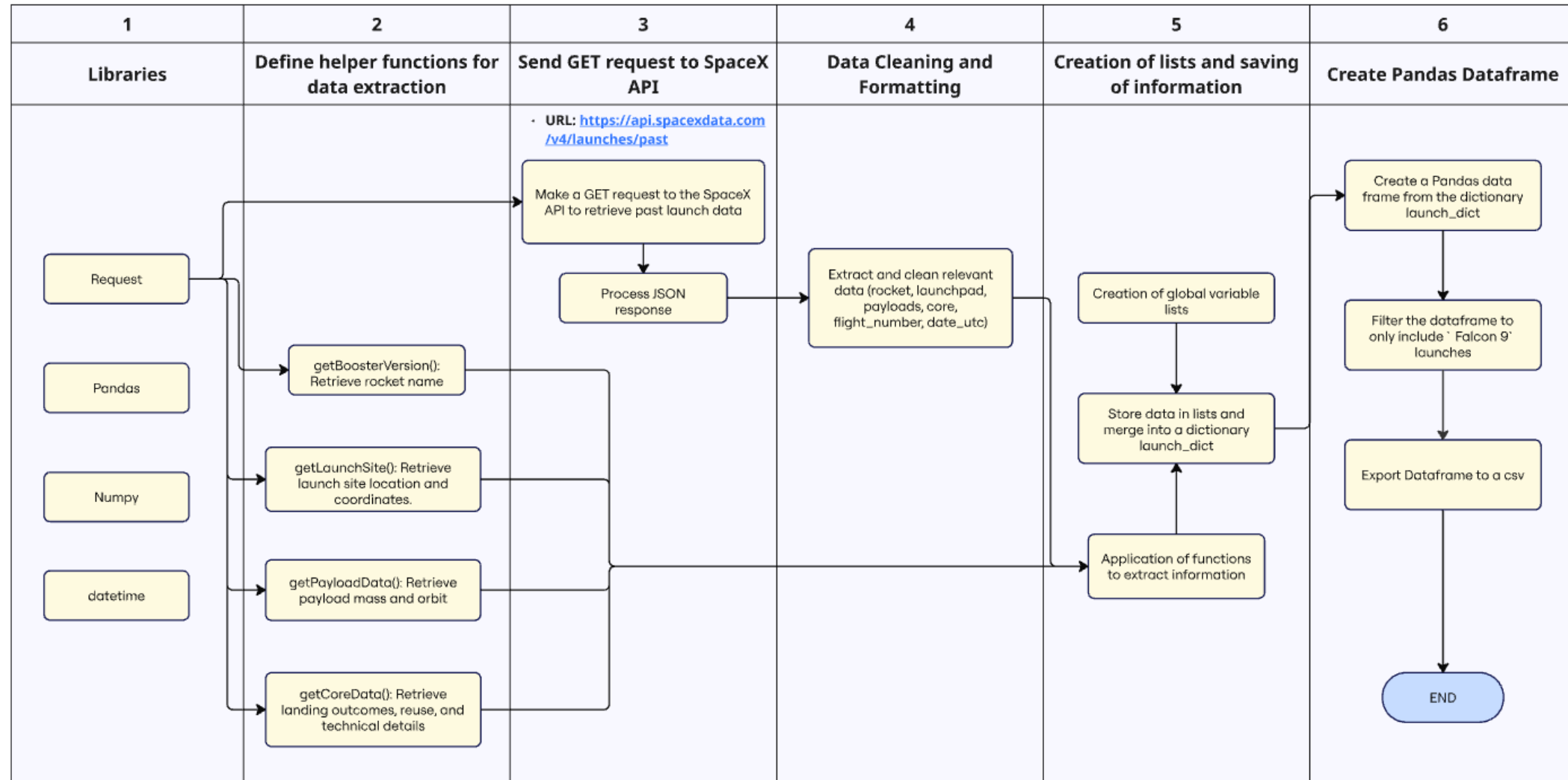
- **Problems you want to find answers**

- **Launch Cost Estimation:** Since reuse of the first stage directly affects pricing, it is necessary to predict whether this stage will be recovered.
- **Successful Landing Prediction:** Instead of applying aerospace physics, the proposal is to train a machine learning model using publicly available data to anticipate whether SpaceX will reuse the first stage.
- **Business Decision Optimization:** SpaceY requires this prediction to set competitive pricing and make strategic decisions in the space market.

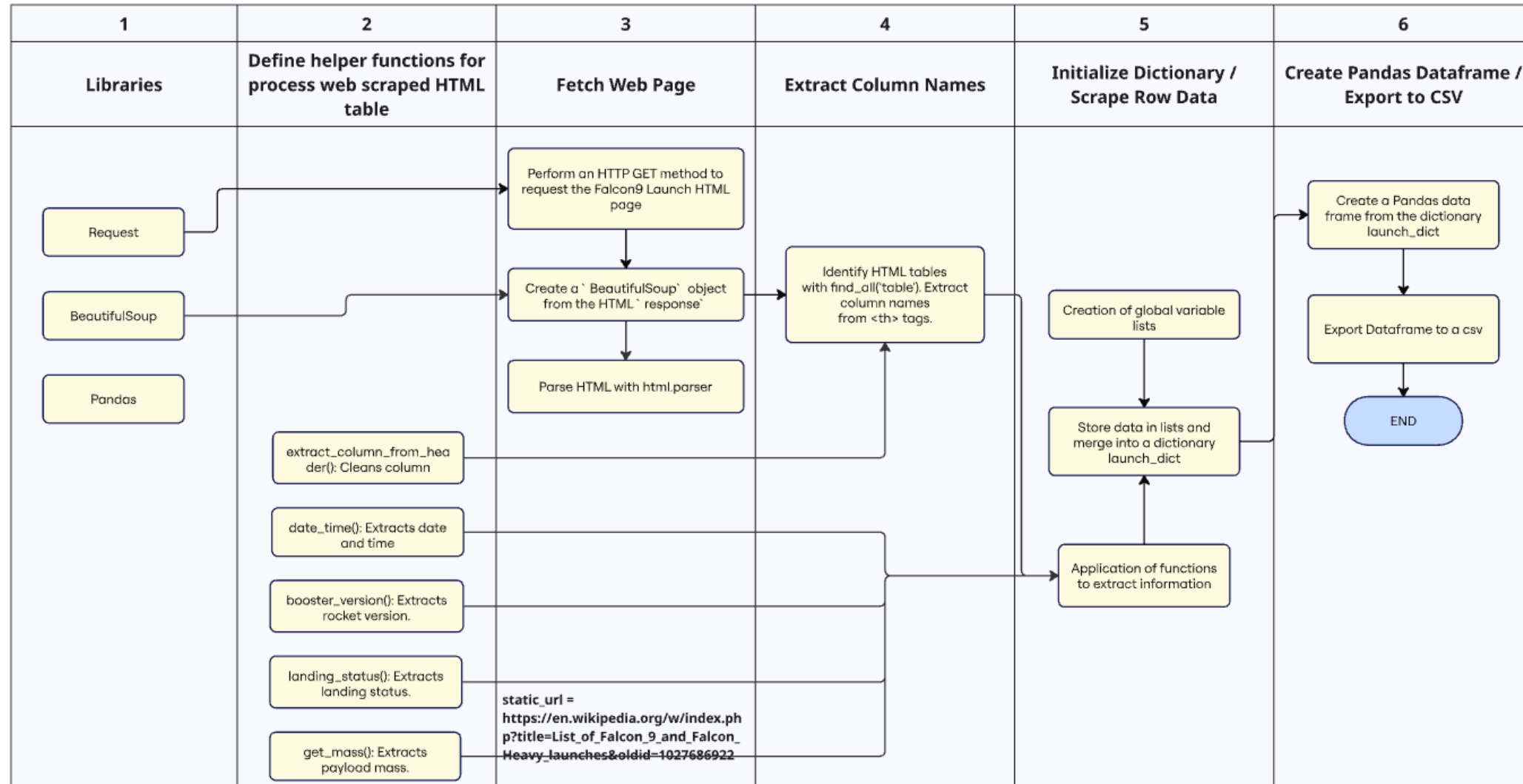
Section 1

Methodology

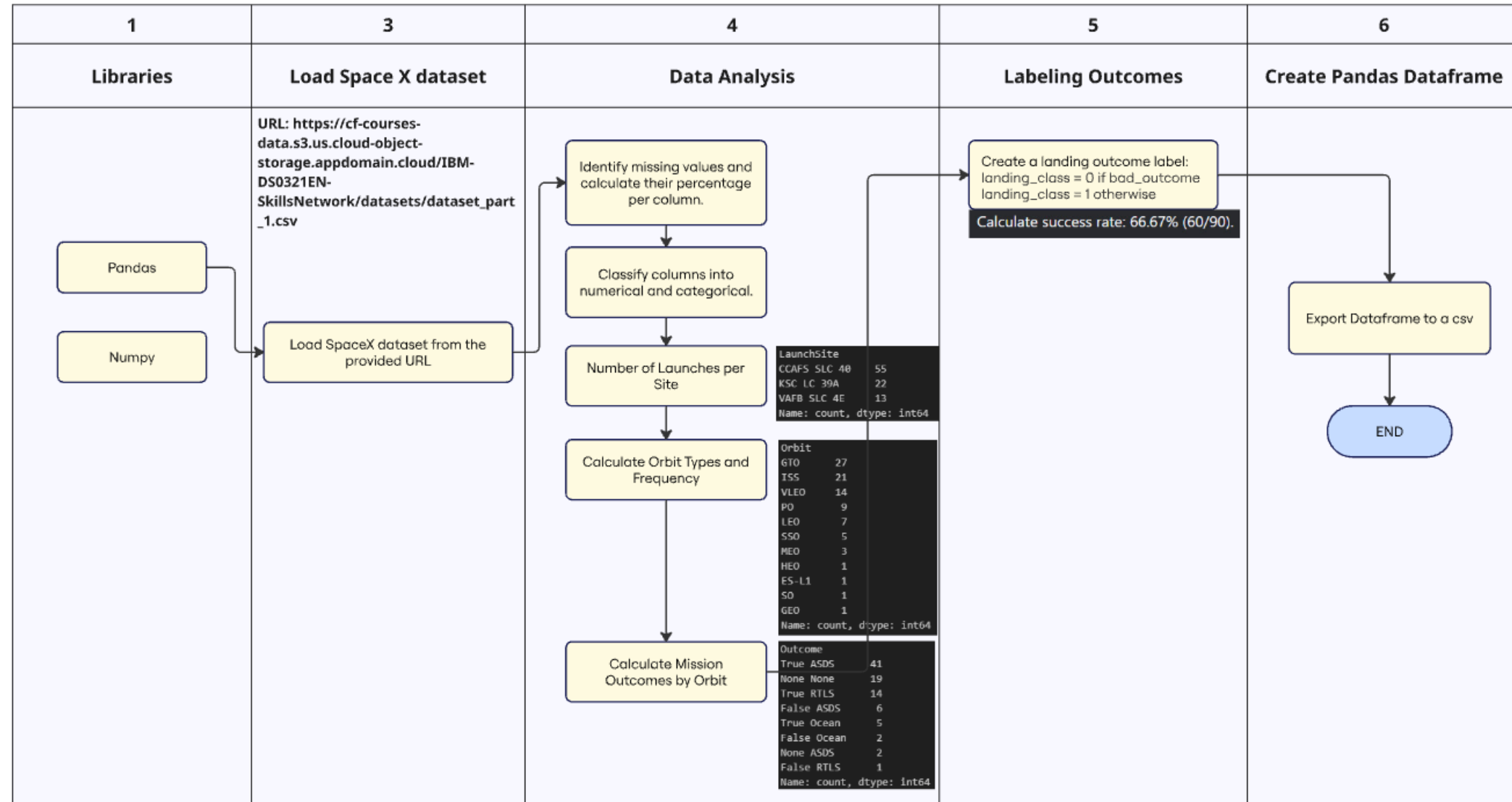
Data Collection – SpaceX API



Data Collection - Scraping



Data Wrangling



EDA with Data Visualization

During the data understanding phase, we generated multiple visualizations, primarily scatter plots, bar charts, and line graphs, with the following objectives:

1. Explore Variable Distribution & Ranges
 - Analyzed key variables (PayloadMass, FlightNumber) to detect outliers, biases, and potential transformations needed for modeling.
2. Identify Key Correlations
 - Used scatter plots to reveal relationships between predictors and the target variable (Class), uncovering patterns such as:
 - Higher success rates with increasing flight numbers (experience-driven improvement).
 - Lower success rates with heavier payloads (technical limitations).
3. Assess Categorical Variables
 - Evaluated the impact of LaunchSite and Orbit on outcomes, linking operational results to geographical or technical constraints.
4. Detect Multivariable Interactions
 - Investigated combined effects (LaunchSite, PayloadMass, Orbit), to inform multivariate modeling strategies.
5. Track Temporal Performance
 - Line graphs illustrated year-over-year trends, showing:
 - Early stagnation (2010–2013).
 - Steady improvement (2014–2018).
 - Peak performance (2019) followed by a minor 2020 dip.

Section 2 contains the visualizations and conclusions derived from the exploratory analysis.

EDA with SQL

- Launch Site Distribution
 - Grouped records by Launch_Site and counted total launches.
 - Sorted in descending order to identify the most active launch sites.
- Pattern-Based Filtering of Launch Sites
 - Retrieved rows where Launch_Site starts with 'CCA'.
 - Applied LIMIT to display the top 5 matches.
- Total Payload for NASA (CRS)
 - Calculated the sum of PAYLOAD_MASS__KG_ for launches with the customer 'NASA (CRS)'.
- Average Payload by Booster Version
 - Computed average payload mass for booster version 'F9 v1.1' using AVG().
- Earliest Launch with Successful Ground Pad Landing
 - Retrieved the minimum Date for records with landing outcome 'Success (ground pad)'.
- Filtered Analysis by Landing Outcome and Payload Range
 - Selected launches that landed on drone ships successfully.
 - Included only records with payload mass between 4000 and 6000 kg.
- Mission Outcome Distribution
 - Grouped data by Mission_Outcome.
 - Counted the number of occurrences for each outcome.
- Booster Version Linked to Maximum Payload
 - Used a subquery to identify the highest payload.
 - Selected the corresponding Booster_Version.
- Drone Ship Failures in 2015
 - Extracted month from Date using substr.
 - Filtered results by year '2015' and failure outcome 'Failure (drone ship)'.
- Landing Outcomes Between 2010 and 2017
 - Filtered launches within a specific date range
 - Grouped and counted Landing_Outcome values
 - Sorted results by descending frequency

Build an Interactive Map with Folium

1. Markers

- Description: Markers were used to pinpoint the locations of the CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, and VAFB SLC-4E launch sites. Each bookmark included a label with the name of the site.
- Purpose
 - Visually identify the exact coordinates of each launch site.
 - Provide additional information by clicking on them (popups).
 - Differentiate between successful (green) and failed (red) launches using colors and symbols.
 - Show the calculation of distances between the launch site and a specific point (coast, trains, cities, etc.)

2. Circles

- Description: Circles were drawn around each launch site with a radius of 700 meters and a semi-transparent blue color.
- Purpose:
 - Highlight the approximate area of each launch site.
 - Display geographic proximity between nearby sites (such as CCAFS LC-40 and CCAFS SLC-40).
 - Improve map readability by grouping related information.

3. Marker Grouping (MarkerCluster)

- Description: The MarkerCluster class was used to group markers close to each other, especially useful for sites with multiple launches (such as CCAFS SLC-40).
- Purpose:
 - Avoid visual clutter when there are many markers in a small area.
 - Allow you to explore launch data by zooming in on specific areas.
 - Show cluster launch counts for an overview.

Build an Interactive Map with Folium

4. Marking coordinates on the map (MousePosition)
 - Description: The MousePosition allows you to obtain the coordinates by hovering the cursor over a point on the map.
 - Purpose
 - Obtain the coordinates (latitude and longitude) to calculate the distances between the launch site and a specific point.
5. Create objects as lines with Folium.PolyLine
 - Description: with Folium.PolyLine you can make a line that represents the distance between the launch point and a specific place.

Build a Dashboard with Plotly Dash

1. Pie Chart

1. Purpose: Display the distribution of successful and failed launches.
2. Functionality:
 1. If "ALL Sites" is selected, shows the proportion of launches by site.
 2. If a specific site is selected, shows success (class=1) vs. failure (class=0) for that site.

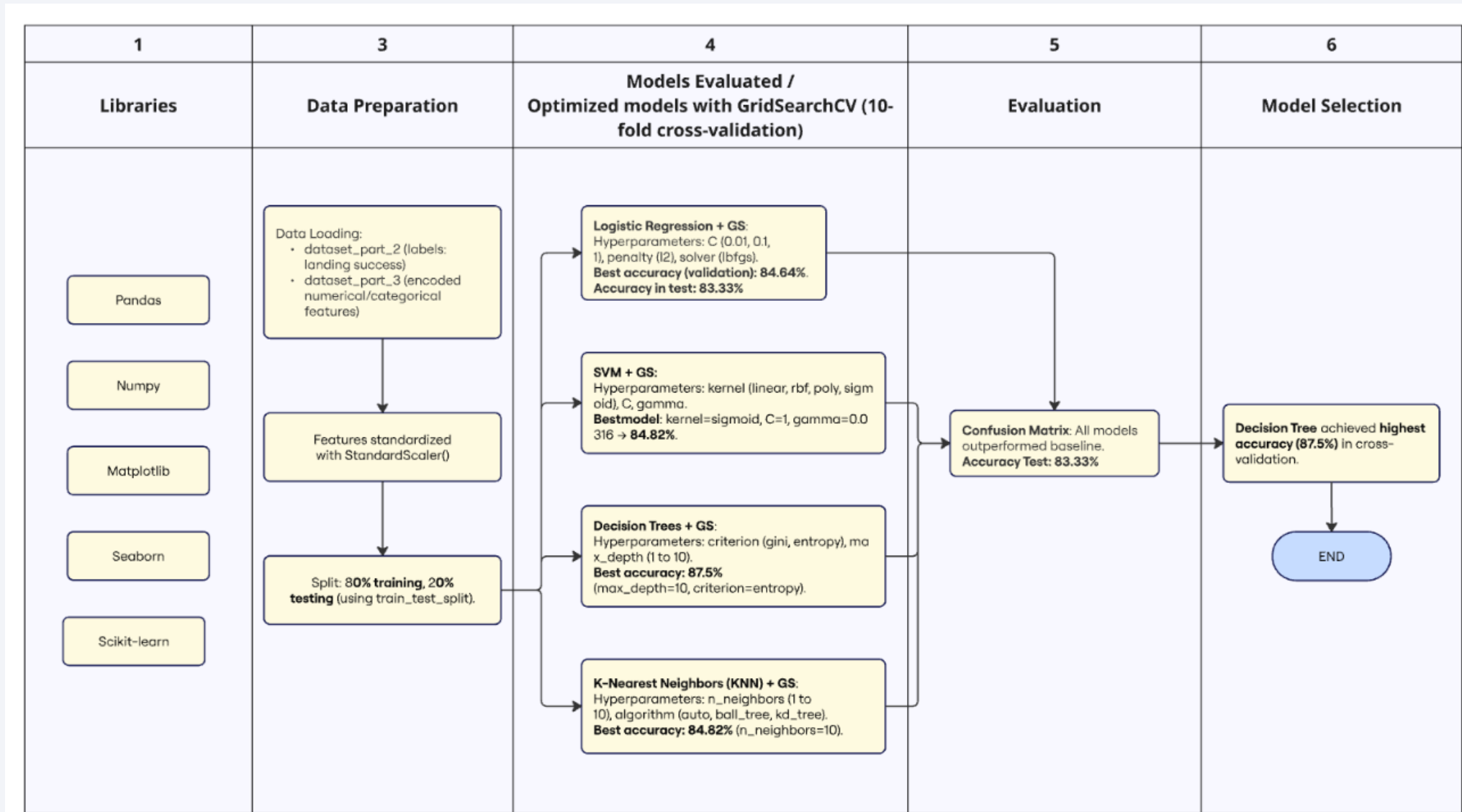
2. Scatter Plot

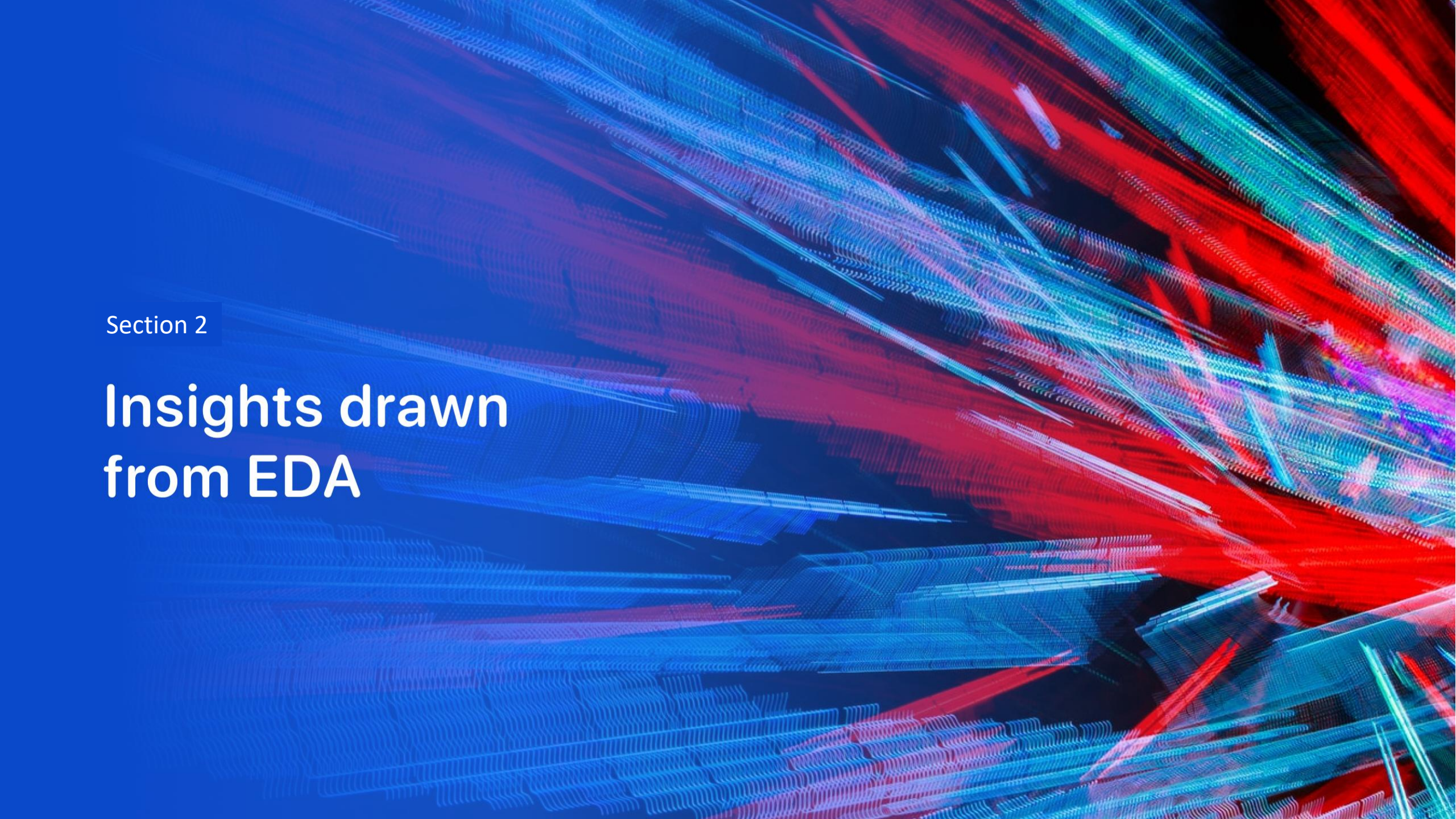
1. Purpose: Visualize the correlation between payload mass and launch success.
2. Functionality:
 1. Filters by payload range (slider) and launch site (dropdown).
 2. Points are colored by booster version category.

3. Interactive Controls:

1. Dropdown: Selects a launch site or "ALL".
2. Slider: Adjusts payload mass range (0 to 10,000 kg).

Predictive Analysis (Classification)

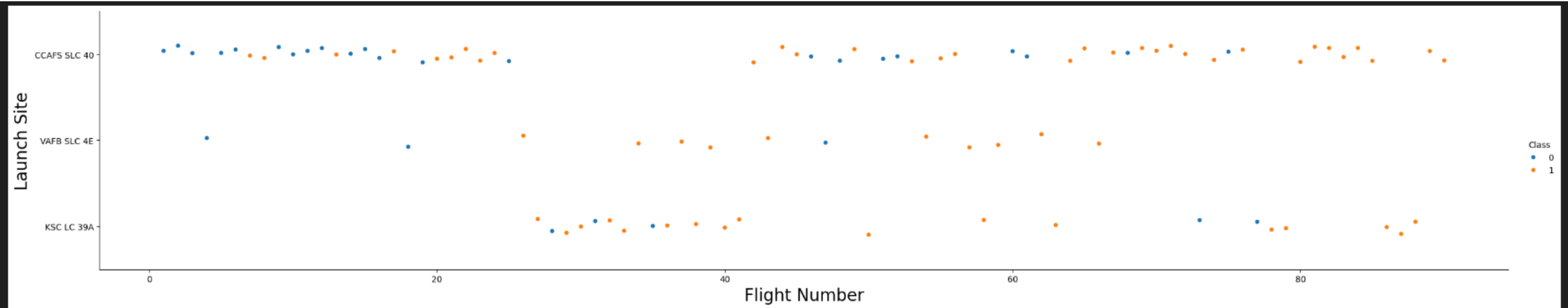


The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

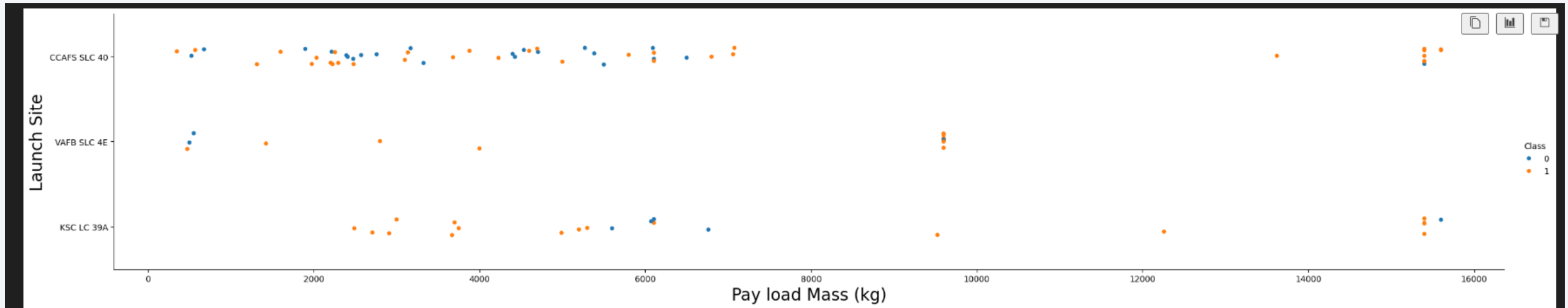


Now try to explain the patterns you found in the Flight Number vs. Launch Site scatter point plots.

KSC LC 39A: This site dominates with a high concentration of successful landings, even from relatively low flight numbers. CCAFS SLC 40: Shows a more balanced mix of successes and failures, though successes increase with flight numbers. VAFB SLC 4E: Has fewer flights and a more scattered distribution, with a lower prevalence of successful landings.

Conclusion: The graph suggests that the launch site significantly influences the first-stage landing outcome, with KSC LC 39A potentially offering better technical, logistical, or environmental conditions for success.

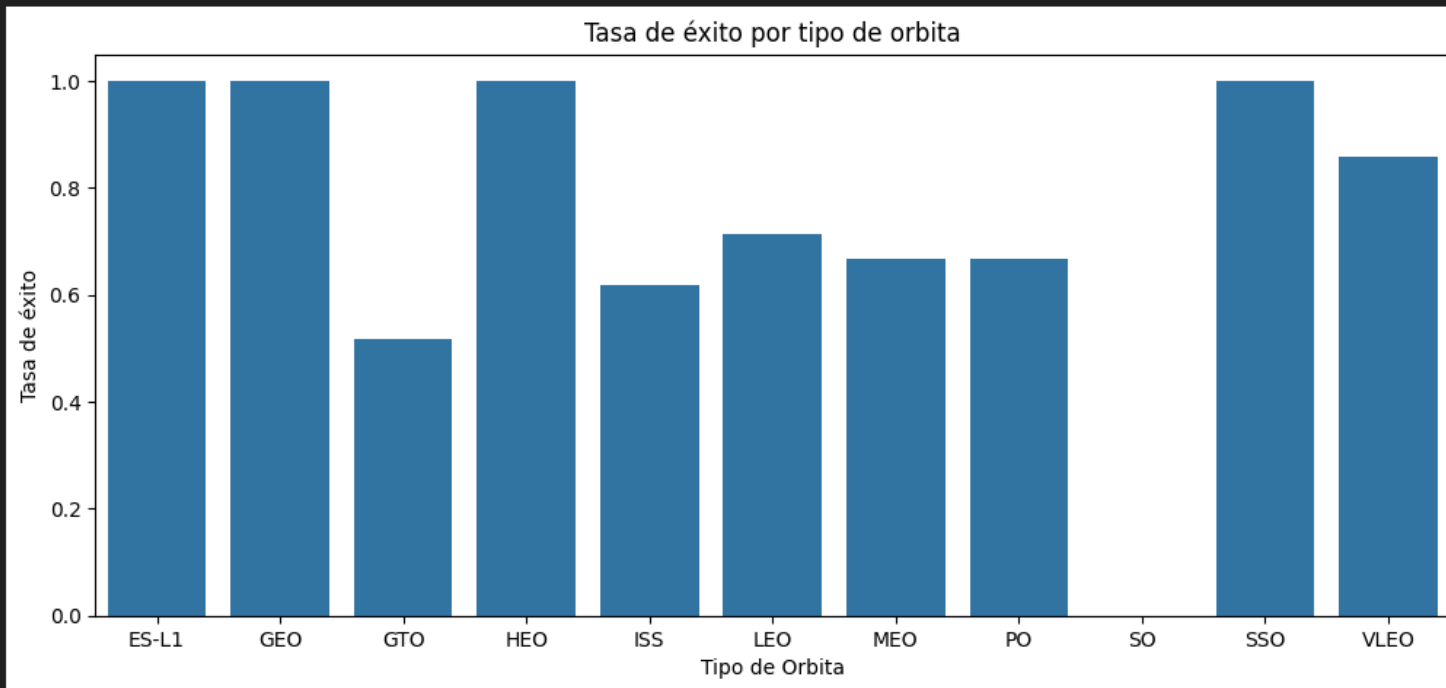
Payload vs. Launch Site



Now if you observe Payload Vs. Launch Site scatter point chart you will find for the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000).

- KSC LC 39A: Shows high variability in payload mass, with launches reaching up to 16,000 kg. Despite this, there is a notable proportion of successes, suggesting this site handles heavy payloads more effectively.
- CCAFS SLC 40 and VAFB SLC 4E: Primarily operate with payloads under 6,000 kg. Both show a mix of successes and failures, though without the operational robustness seen at KSC LC 39A.
- Success does not appear to depend solely on payload weight but rather on how the launch site manages this variable. KSC LC 39A demonstrates greater technical resilience when handling demanding payloads.

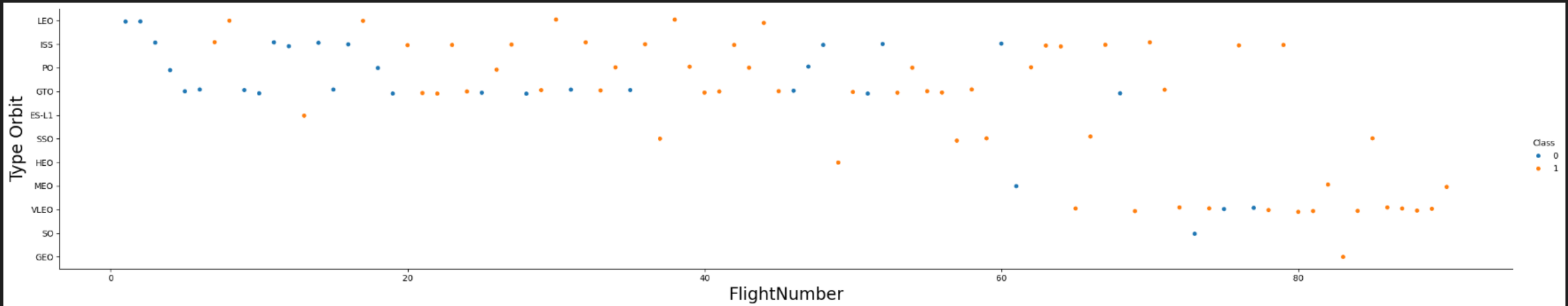
Success Rate vs. Orbit Type



Analyze the plotted bar chart try to find which orbits have high sucess rate.

- ES-L1, GEO, HEO, SSO: All launches to these orbits were successful, suggesting high reliability—likely linked to better planning or technological maturity. However, except for SSO, each has only one recorded launch.
- GTO Orbit: Shows the lowest success rate (~52%). As a transfer orbit, it may involve additional complexities, post-launch maneuvers or reliance on secondary stages, increasing risk.
- Medium Orbits (ISS, LEO, MEO, PO): Success rates range between 60% and 80%, reflecting persistent technical or consistency challenges.
- VLEO Orbit: Maintains a high success rate (~86%), indicating stable operations in extremely low orbits.

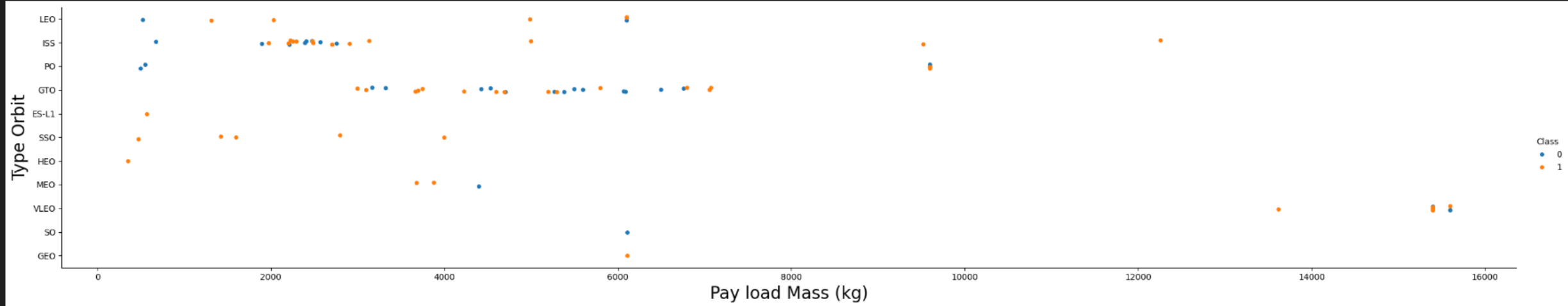
Flight Number vs. Orbit Type



You should see that in the LEO orbit the Success appears related to the number of flights; on the other hand, there seems to be no relationship between flight number when in GTO orbit.

- LEO, ISS, and PO Orbits: These show a high density of successful cases as the flight number increases. This suggests that these orbits, in addition to being more common, benefit from accumulated experience and optimized processes.
- GTO Orbit: Consistently associated with failed cases, even in later flights, indicating persistent technical challenges. This may be due to complex post-launch maneuvers or stricter trajectory requirements.
- ES-L1, SSO, GEO, HEO, and VLEO Orbits: Although less frequent, launches to these orbits tend to be successful. The low frequency suggests mission-specific specialization or more rigorous planning.

Payload vs. Orbit Type

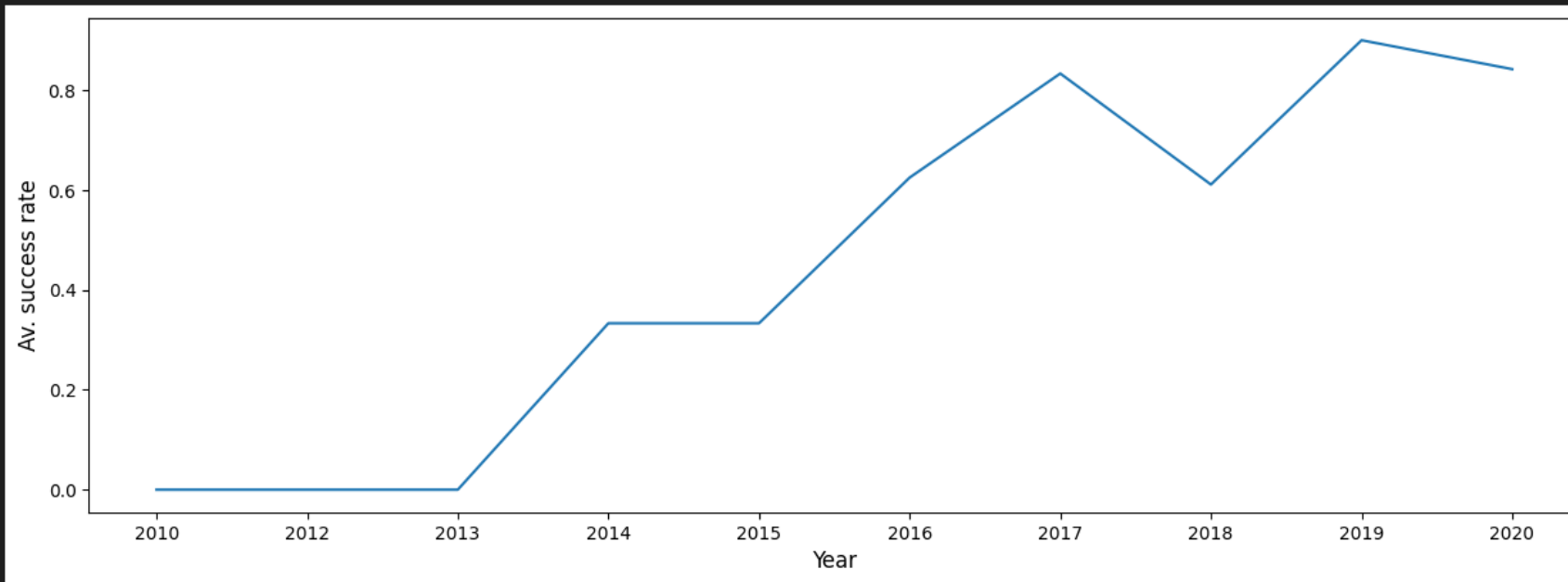


With heavy payloads the successful landing or positive landing rate are more for Polar,LEO and ISS.

However for GTO we cannot distinguish this well as both positive landing rate and negative landing(unsuccesful mission) are both there here.

- Focus on LEO, ISS, PO, and GTO: Most launches target these orbits, suggesting their predominant use for satellite missions, resupply operations, or initial transfer stages.
- LEO and ISS: Typically handle lighter payloads (under 4,000 kg), ideal for observation satellites, tests, or station deliveries.
- GTO and SSO: Exhibit high payload variability (up to 8,000 kg), reflecting more complex missions and orbital transfers.
- GEO and ES-L1: Have lower launch frequencies, indicating mission-specific planning and specialization.

Launch Success Yearly Trend



You can observe that the success rate since 2013 kept increasing till 2017 (stable in 2014) and after 2015 it started increasing.

- Initial Stagnation Phase (2010–2013): The line remains at zero, suggesting launches during this period failed to meet defined "success" criteria. This likely reflects a developmental stage, experimentation, or recurring failures.
- Sustained Growth (2014–2018): From 2014 onward, success rates improve steadily. This upward trend reveals the implementation of technical corrections, accumulated experience, and refined protocols.
- Technical Peak (2019): The maximum value is reached this year, indicating a convergence of operational maturity, optimized technologies, and likely consolidation of reusable methods.
- Minor Setback (2020): Though not critical, the dip suggests potential anomalies, protocol reconfigurations, or external condition effects.

All Launch Site Names

The following section outlines the SQL queries applied during data analysis:

1. A list of unique launch sites. A total of four launch sites were identified, with CCAFS SLC-40 standing out with 34 launches.

```
1 %%sql
2 SELECT
3     Launch_Site,
4     COUNT(*) AS Launch_Count
5 FROM SPACEXTABLE
6 GROUP BY Launch_Site
7 ORDER BY Launch_Count DESC;
```

* [sqlite:///my_data1.db](#)
Done.

Launch_Site	Launch_Count
CCAFS SLC-40	34
CCAFS LC-40	26
KSC LC-39A	25
VAFB SLC-4E	16

Launch Site Names Begin with 'CCA'

2. This SQL query retrieves the first 5 records from the SPACEXTABLE where the launch site (Launch_Site) starts with 'CCA'.

```
1 %sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
```

* [sqlite:///my_data1.db](#)

Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

3. This SQL query calculates the total sum of payload mass (PAYLOAD_MASS__KG_) launched into space for missions where the customer (Customer) is NASA (CRS). R// 45,596 kg.

```
1 %sql SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_NASA_CRS FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

Total_Payload_NASA_CRS

45596

Average Payload Mass by F9 v1.1

4. This SQL query calculates the average payload mass (PAYLOAD_MASS__KG_) for missions where the booster version (Booster_Version) is 'F9 v1.1'.

```
1 %sql SELECT AVG(PAYLOAD_MASS__KG_) AS AVG_Payload_NASA_CRS FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1';
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

AVG_Payload_NASA_CRS

2928.4

First Successful Ground Landing Date

5. This SQL query retrieves the earliest date (MIN(Date)) when a SpaceX rocket successfully landed on a ground pad (Landing_Outcome = 'Success (ground pad)').

```
1 %sql SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';

* sqlite:///my\_data1.db
Done.

MIN(Date)
2015-12-22
```

Successful Drone Ship Landing with Payload between 4000 and 6000

6. This SQL query retrieves the booster version (Booster_Version) and payload mass (PAYLOAD_MASS_KG_) for all missions where:

- The landing was successful on a drone ship.
- The payload mass was between 4,000 kg and 6,000 kg.

```
1 %sql SELECT "Booster_Version", "PAYLOAD_MASS_KG_" FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND "PAYLOAD_MASS_KG_" BETWEEN 4001 AND 5999;
```

* [sqlite:///my_data1.db](#)

Done.

Booster_Version	PAYLOAD_MASS_KG_
F9 FT B1022	4696
F9 FT B1026	4600
F9 FT B1021.2	5300
F9 FT B1031.2	5200

Total Number of Successful and Failure Mission Outcomes

7. This SQL query counts and groups SpaceX mission outcomes (Mission_Outcome) to show how many times each outcome occurred.

```
1 %sql SELECT Mission_Outcome, COUNT(*) AS Outcome_Count FROM SPACEXTABLE GROUP BY Mission_Outcome;
```

* [sqlite:///my_data1.db](#)
Done.

Mission_Outcome	Outcome_Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

8. This SQL query identifies the booster version (Booster_Version) that carried the heaviest payload recorded in the SPACEXTABLE.

```
1 %sql SELECT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS_KG_" = (SELECT MAX(PAYLOAD_MASS_KG_) FROM SPACEXTABLE);
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

Booster_Version

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

2015 Launch Records

9. This SQL query retrieves information about failed drone ship landings (Failure (drone ship)) during 2015, displaying the month, landing outcome, booster version, and launch site.

```
1  %%sql
2  SELECT
3  substr(Date, 6, 2) AS Month,
4  Landing_Outcome,
5  Booster_Version,
6  Launch_Site
7  FROM SPACEXTABLE
8  WHERE
9  substr(Date, 0, 5) = '2015'
10 AND Landing_Outcome = 'Failure (drone ship)';
```

* [sqlite:///my_data1.db](#)

Done.

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

10. This SQL query counts and categorizes landing outcomes (Landing_Outcome) of SpaceX missions between June 4, 2010, and March 20, 2017, showing how many times each outcome occurred, sorted from most to least frequent.

```
1  %%sql
2  SELECT
3      Landing_Outcome,
4      Date,
5      COUNT(*) AS Outcome_Count
6  FROM SPACEXTABLE
7  WHERE
8      Date BETWEEN '2010-06-04' AND '2017-03-20'
9  GROUP BY Landing_Outcome
10 ORDER BY Outcome_Count DESC;
```

* [sqlite:///my_data1.db](#)

Done.

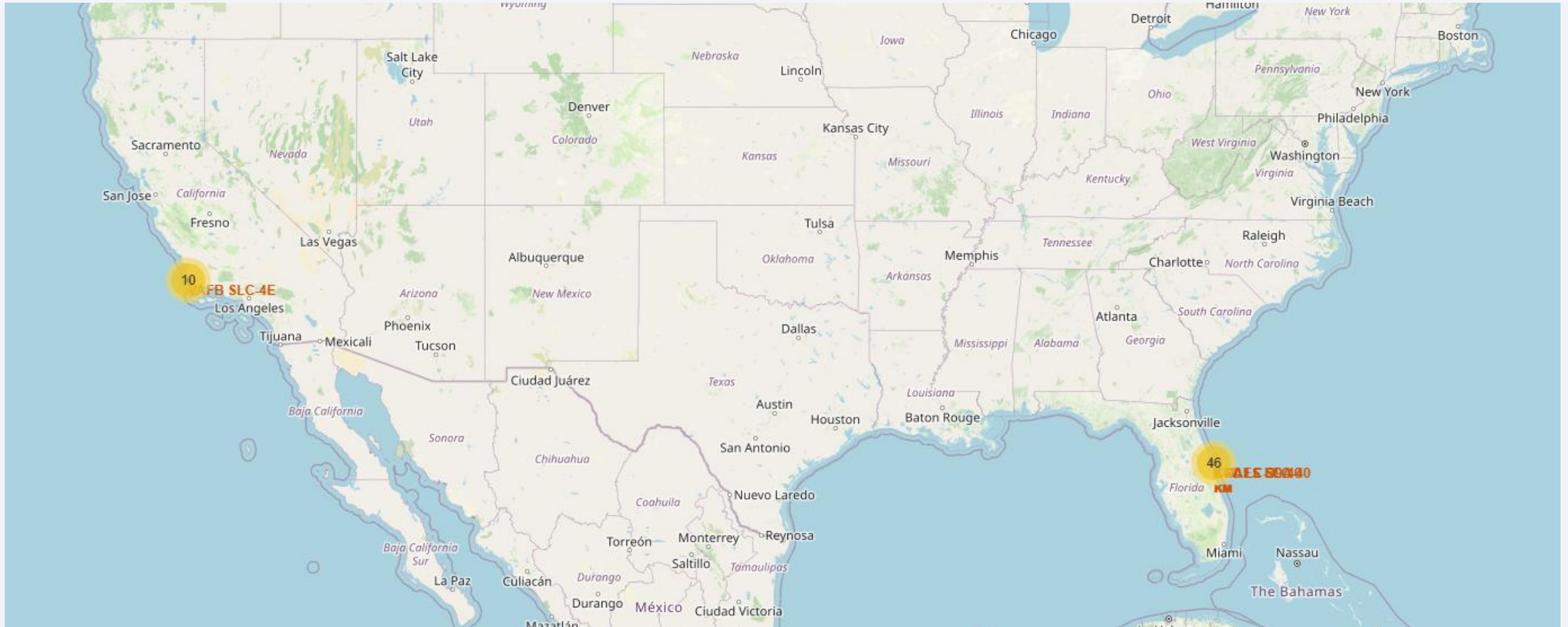
Landing_Outcome	Date	Outcome_Count
No attempt	2012-05-22	10
Success (drone ship)	2016-04-08	5
Failure (drone ship)	2015-01-10	5
Success (ground pad)	2015-12-22	3
Controlled (ocean)	2014-04-18	3
Uncontrolled (ocean)	2013-09-29	2
Failure (parachute)	2010-06-04	2
Precluded (drone ship)	2015-06-28	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

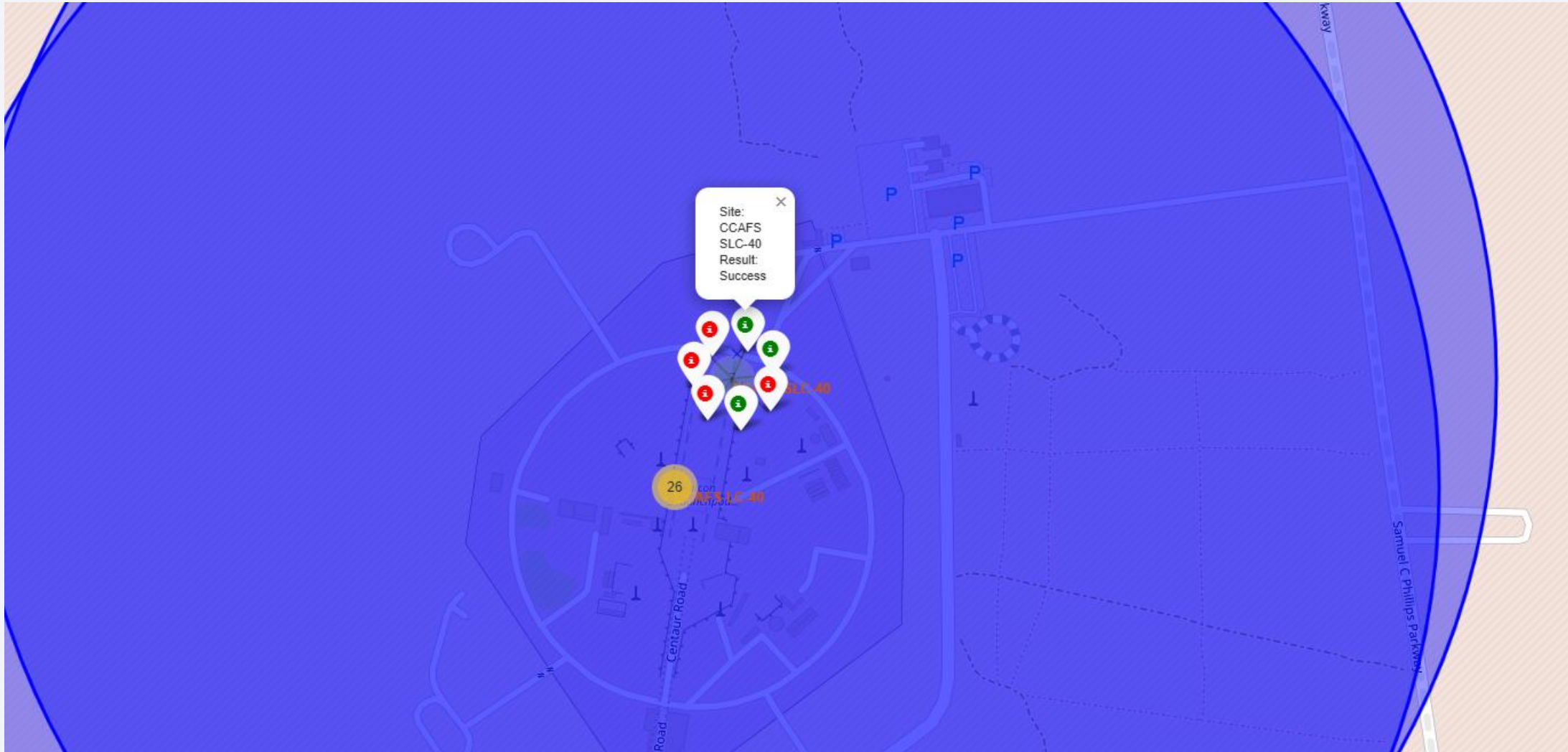
Section 3

Launch Sites Proximities Analysis

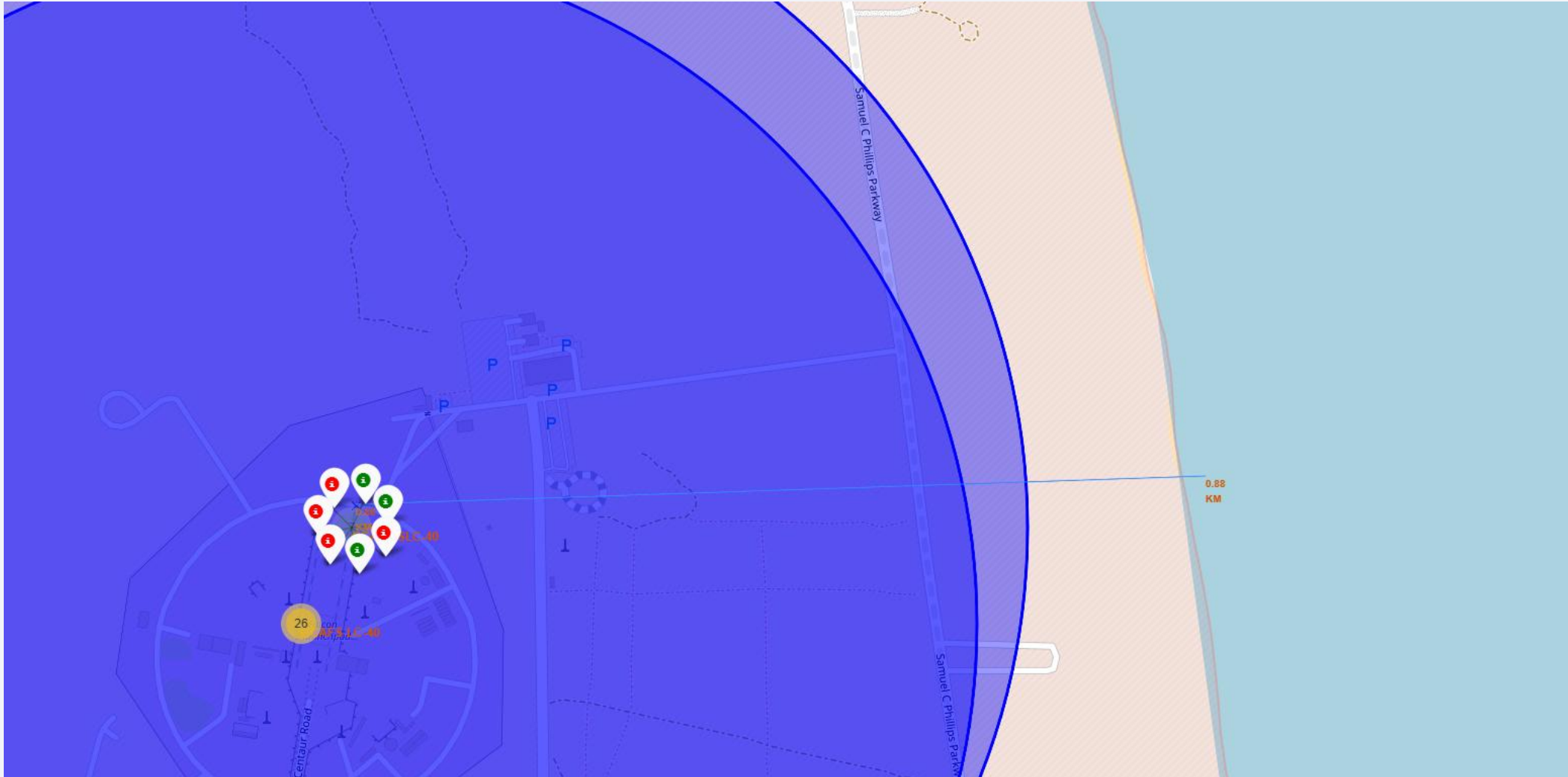
Launch site location markers on the global map



Color-labeled launch outcomes on the map



Launch site map and the distance to the coastline



Interactive Map of SpaceX Launches: Sites and Outcomes (Success/Failure) - Conclusions

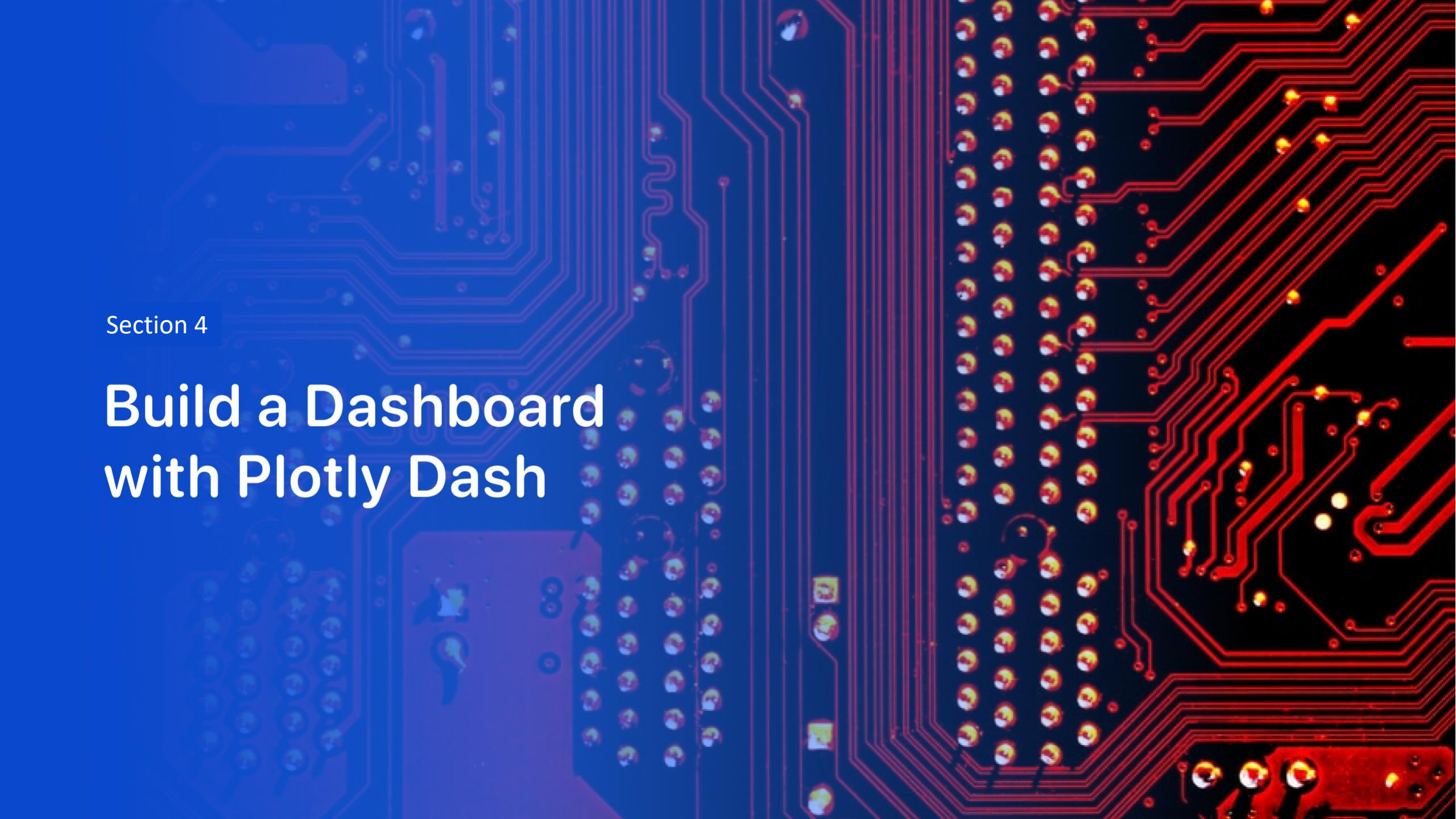
Based on the interactive spatial analysis of SpaceX's launch sites using Folium, with a focus on the relationship between location and launch success, the following conclusions can be drawn:

1. Geographical distribution of launch sites:

- Proximity to the coast: All sites (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A in Florida, and VAFB SLC-4E in California) are located near the coast. This suggests that SpaceX prioritizes coastal locations, likely because of:
 - Safety: Minimize risks to terrestrial populations.
 - Launch routes: Facilitate trajectories over the ocean for expendable rocket stages.
- Nearness to the equator: Sites in Florida (latitude $\sim 28^\circ\text{N}$) are closer to the equator than Vandenberg ($\sim 34^\circ\text{N}$). This is critical for launches into equatorial orbits, as they take advantage of the Earth's rotational velocity (higher at low latitudes).

2. Success rate per site

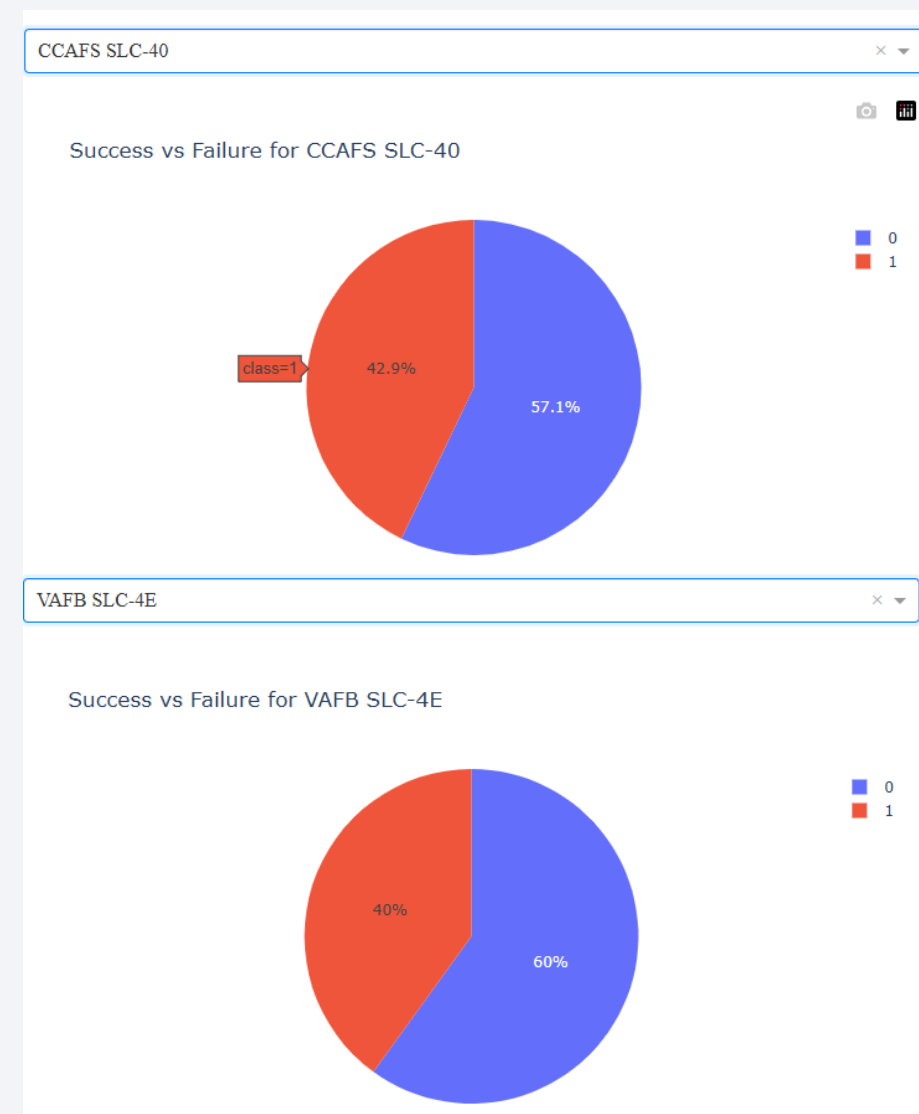
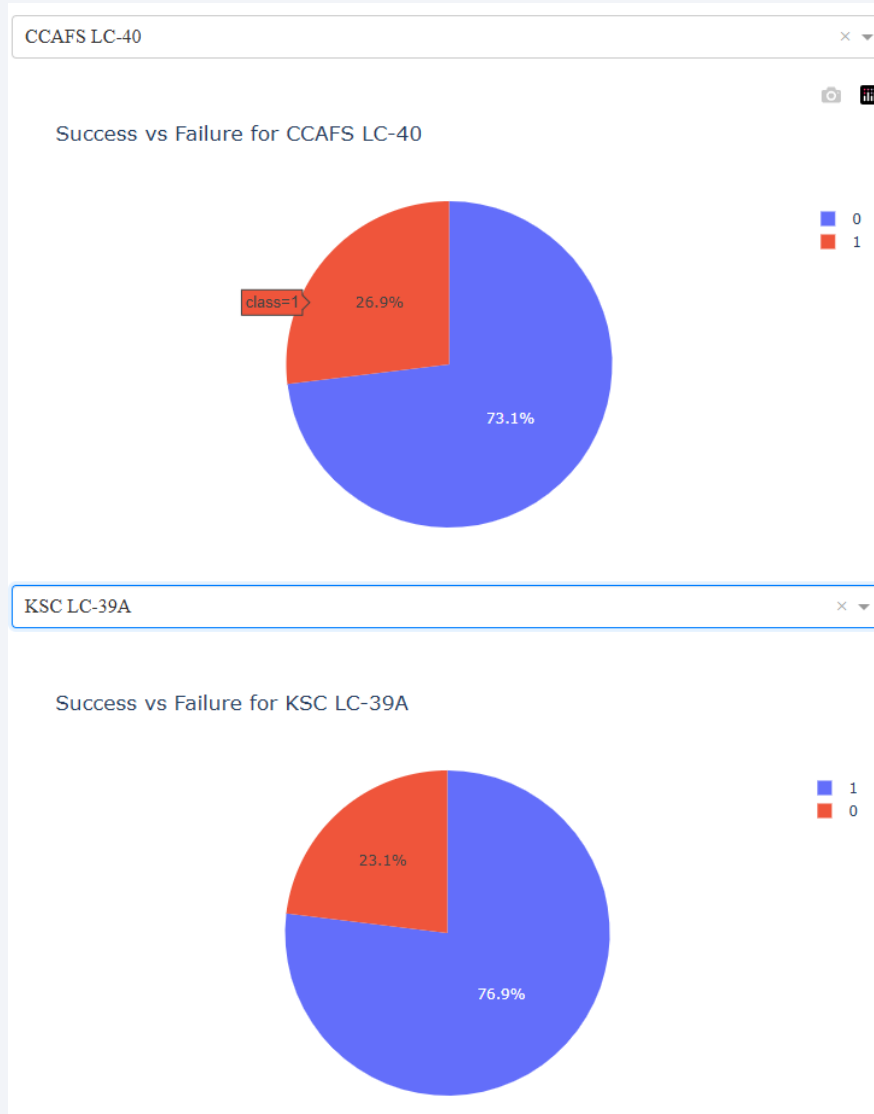
- KSC LC-39A has the highest success rate (76.9%), followed by CCAFS SLC-40 (42.9%) and VAFB SLC-4E (40%). CCAFS LC-40 has the lowest rate (26.9%).
- Possible reasons:
 - Technical Improvements: KSC LC-39A is a NASA Historic Site, adapted for critical missions.
 - Type of missions: Some sites may be used for payloads or more complex orbits (e.g. polar orbits from Vandenberg).
 - Age: CCAFS LC-40 was the first to be used (2010), which may have involved early learning curves.



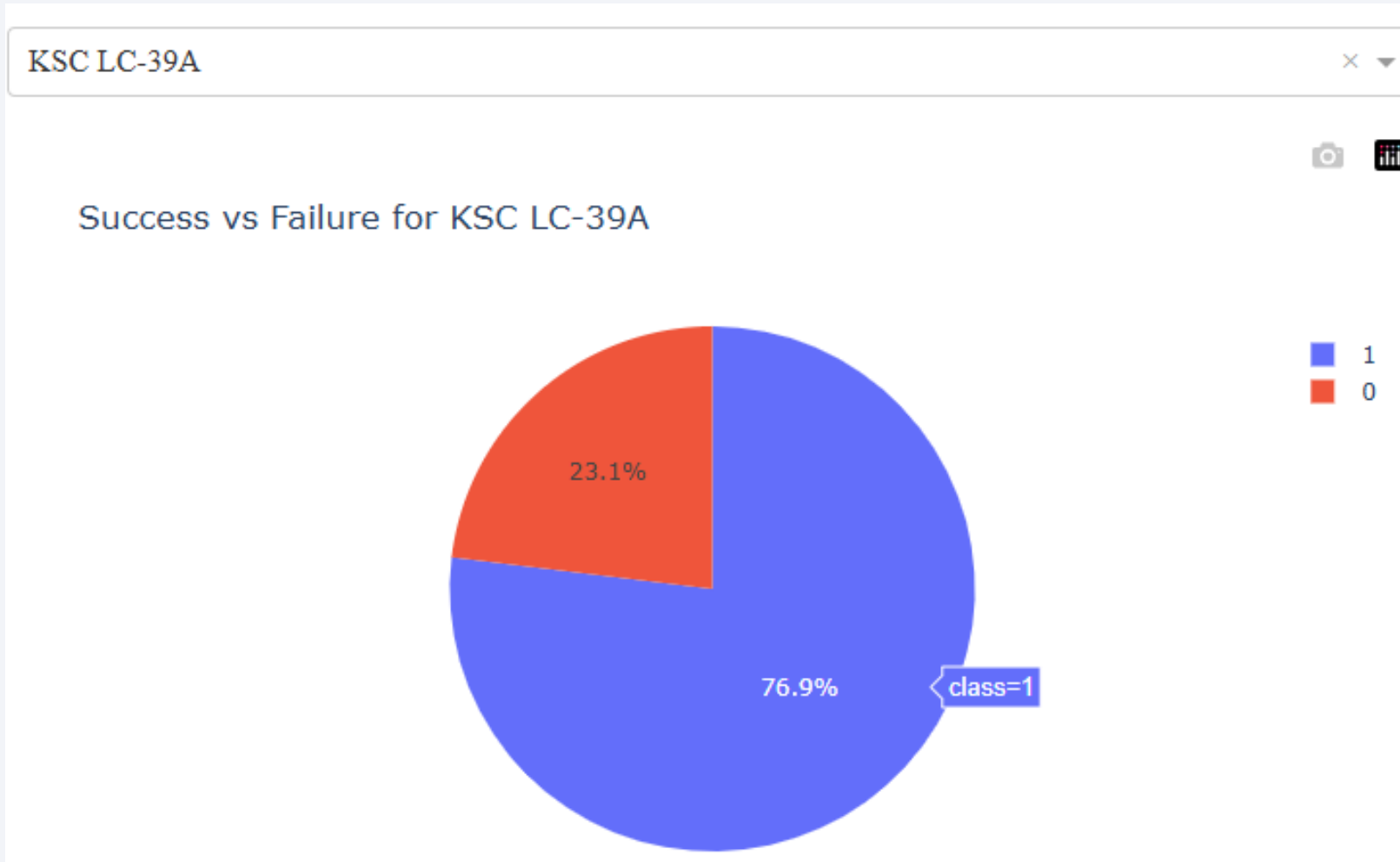
Section 4

Build a Dashboard with Plotly Dash

Pie chart of the success rate by launch sit



Piechart for the launch site with highest launch success ratio



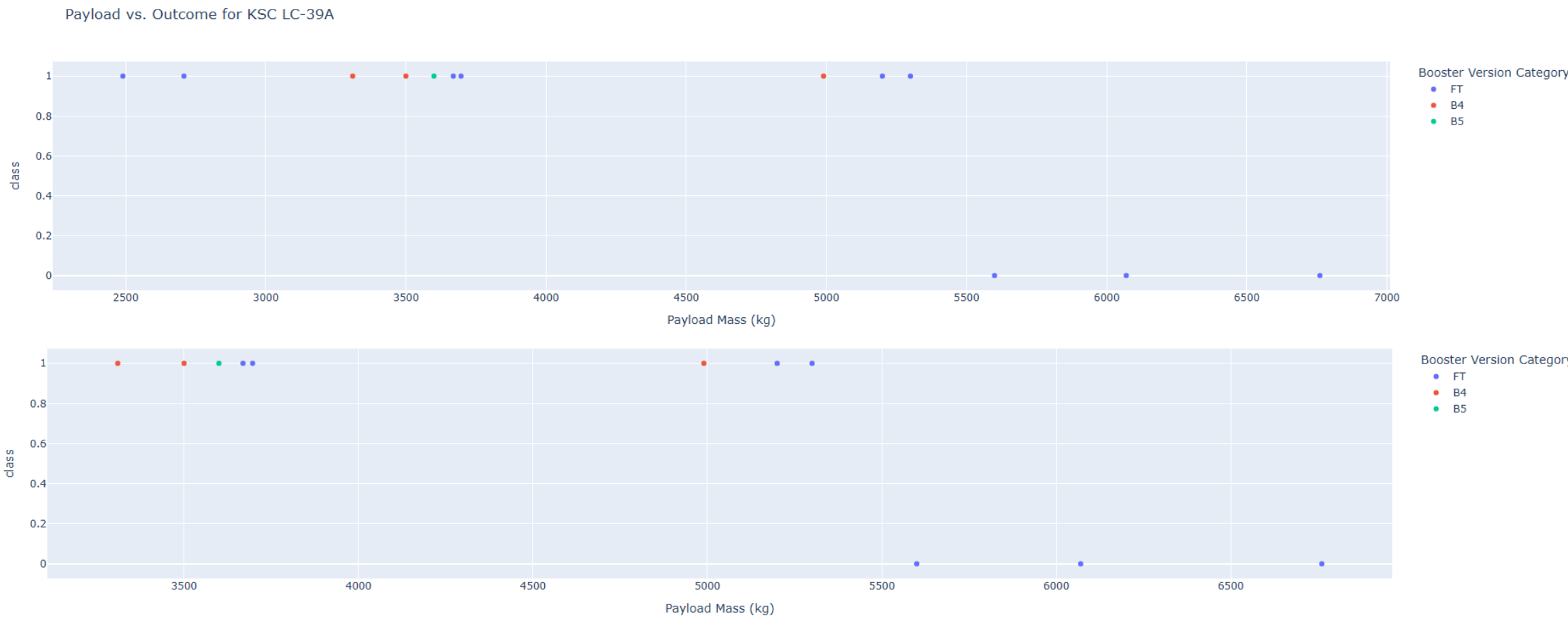
Scatter plot of payload mass versus launch success by booster version



Scatter plot of payload mass versus launch success by booster version



Scatter plot of payload mass versus launch success by booster version



Scatter plot of payload mass versus launch success by booster version



Scatter plot of payload mass versus launch success by booster version - Conclusions

1. Success Rate by Launch Site:

- The pie chart indicates that some sites, like KSC LC-39A, have a higher success rate, possibly due to better infrastructure or operational efficiency.
- Sites like VAFB SLC-4E may underperform, suggesting technical or environmental challenges.

2. Payload Mass vs. Success:

- The scatter plot shows no strong correlation between payload mass and launch success. Success/failure is spread across all mass ranges.
- However, certain booster versions (e.g., Falcon 9 B5) might perform better at specific payload ranges.

3. Booster Version Matters:

- The Booster Version Category is a critical factor. Some versions consistently succeed, while others are less reliable.

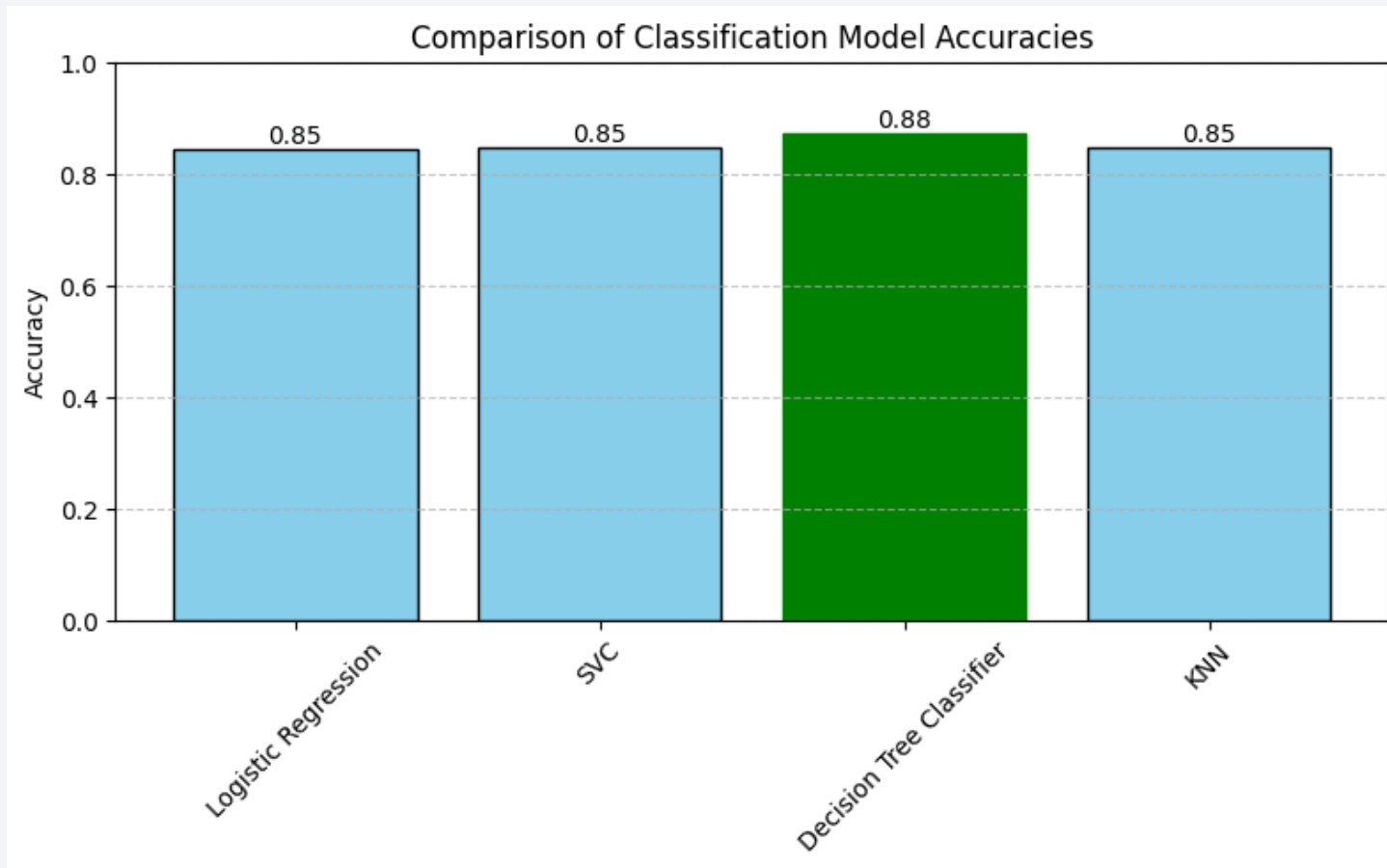
4. Site-Specific Trends:

- Filtering by site reveals stark differences. For example, CCAFS SLC-40 might have a balanced success/failure ratio, whereas KSC LC-39A excels.

Section 5

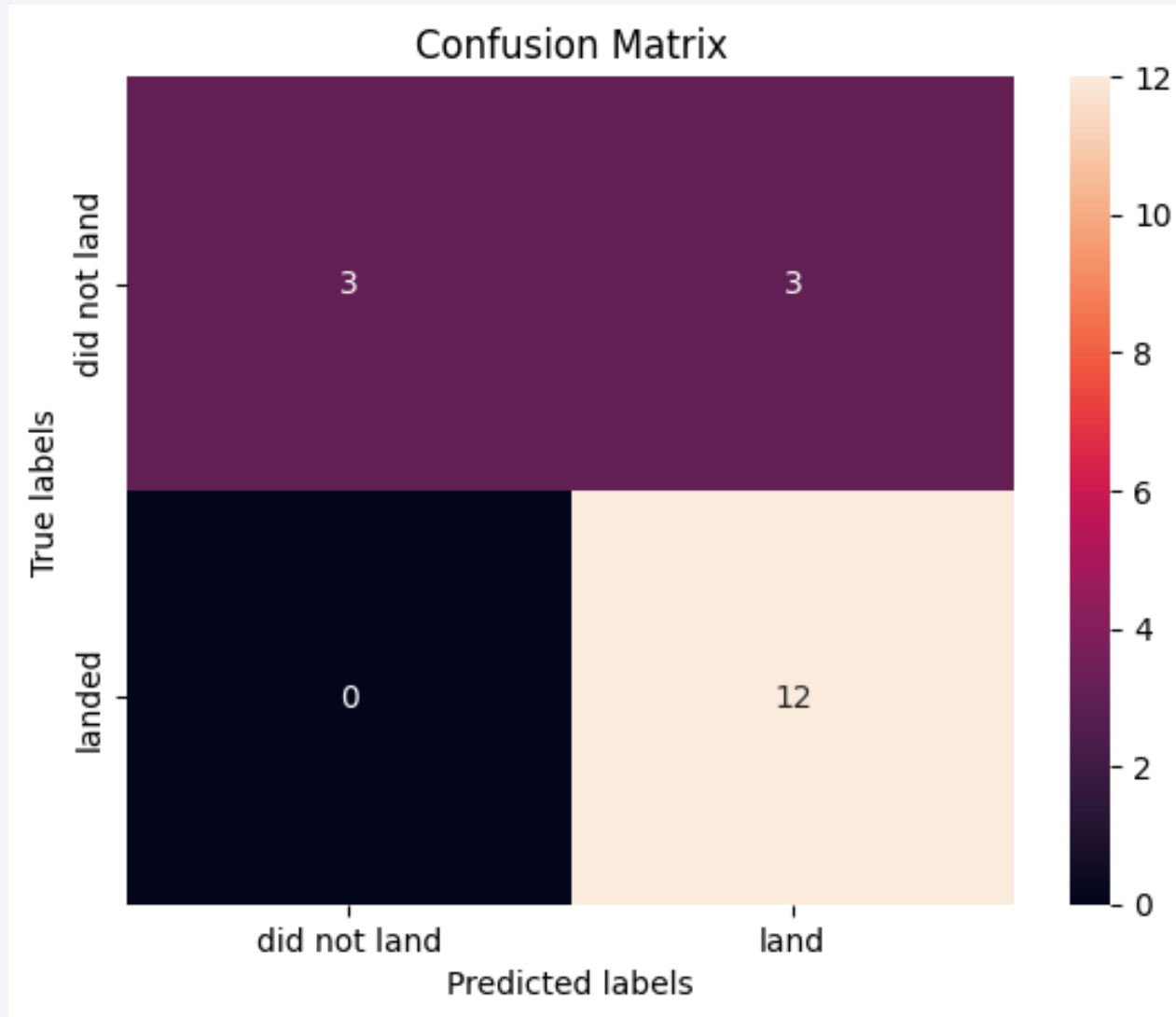
Predictive Analysis (Classification)

Classification Accuracy



Based on cross-validation results, the reported accuracies represent consistent, generalizable performance estimates. The Decision Tree Classifier achieved the highest average accuracy (88%), slightly outperforming Logistic Regression, SVC, and KNN (85%). This suggests the tree adapted best to the data across folds, though it's important to monitor for overfitting.

Confusion Matrix



The confusion matrix shows that the Decision Tree model effectively distinguishes between successful and failed rocket launches, with a high count of true positives and true negatives. This means the model correctly identifies both launch outcomes. The low occurrence of false predictions suggests strong discriminative performance.

Conclusions

1. Impact of Launch Site on Success

- KSC LC-39A demonstrated the highest success rate (76.9%), highlighting its technical and logistical superiority, likely due to its legacy as a historic NASA site.
- CCAFS SLC-40 and VAFB SLC-4E showed mixed results, suggesting operational challenges or limitations in specific missions (e.g., polar orbits).
- Proximity to the coast and the equator (e.g., Florida) appears to be a critical factor for safety and launch efficiency.

2. Payload Mass vs. Launch Success

- No clear correlation was found between payload mass and launch success. However, sites like KSC LC-39A handled heavier payloads (up to 16,000 kg) more effectively, indicating technical resilience.
- Orbits such as GTO had lower success rates (~52%), possibly due to post-launch maneuver complexities.

3. Temporal Trends

- 2010-2013: Experimental phase with high failure rates.
- 2014-2019: Sustained improvement, peaking in 2019 due to advancements in rocket reusability.
- 2020: Minor setback, potentially due to technical adjustments or external factors.

4. Predictive Models

- The Decision Tree was the most accurate model (87.8% accuracy), outperforming Logistic Regression and SVM.
- The confusion matrix confirmed its ability to distinguish between successful and failed landings, with few false positives/negatives.

Conclusions

5. Analytical Tools

- Interactive visualizations (Folium, Plotly Dash) helped identify geographic patterns and key correlations.
- SQL enabled historical trend analysis, such as total payload mass for NASA (45,596 kg) and the first successful ground landing (December 2015).

6. Recommendations for SpaceY

- Prioritize launch sites near the equator with proven infrastructure (e.g., KSC LC-39A).
- Invest in rocket versions with higher success rates (e.g., Falcon 9 B5).
- Monitor complex orbits (e.g., GTO) to mitigate risks.

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!

